Moving-Prime Arithmetic and a Kummer-Period Bridge for the Alaoglu–Erdős Problem

A nonduplicative progress report with exact geometric p-orderings, a sharp half-mass barrier, a gcd-compatibility endgame, and a motivic conditional proof

OpenAI GPT-5.6 Pro

22 August 2026

Abstract

Let $x \in \mathbb{R}$ and suppose that $M = 2^x$ and $N = 3^x$ are integers. The Alaoglu–Erdős conjecture asserts that x is an integer. This report does not claim an unconditional proof. It records a set of new results and proof targets that were selected specifically to avoid reproducing the broad determinant, Mahler-function, fixed-prime p-adic, interpolation, approximation-theoretic, and model-theoretic developments of the supplied Report A.

The main unconditional result is an exact simultaneous p-ordering theorem for every integral geometric progression $\{aq^n : n \geq 0\}$. The natural ordering works for every prime at once, and its generalized factorial is

$$a^k q^{\binom{k}{2}} \prod_{r=1}^{k} (q^r - 1).$$

This gives a canonical Newton–Gaussian basis for all rational polynomials integer-valued on the progression. An exact local valuation formula then splits the logarithmic denominator mass into two asymptotically equal parts: one half is concentrated at the fixed primes dividing q, while the other half is carried by the cyclotomic factors $q^r - 1$. On the full geometric Vandermonde scale the same phenomenon occurs at cubic order. Moreover, the cyclotomic half escapes every fixed finite set of primes. Consequently, a fixed-2/3-adic determinant argument can capture at most one half of the universal geometric denominator at leading order; the missing arithmetic must move with the interpolation scale.

To turn this diagnosis into an endgame, the report introduces a weighted moving-prime compatibility mass

$$\mathfrak{C}_d(q, Q) = \sum_{r=1}^{d} (d + 1 - r) \log \gcd(q^r - 1, Q^r - 1).$$

A theorem of Bugeaud–Corvaja–Zannier implies a sharp zero–one law when q is prime: $\mathfrak{C}_d(q, Q)$ has full cubic mass if Q is an integral power of q, and is $o(d^3)$ if q and Q are multiplicatively independent. Therefore the original conjecture follows from one concrete bridge statement: a common real exponent for $(2, M)$ and $(3, N)$ must force a positive proportion of moving-prime compatibility in at least one pair. A block-Vandermonde lemma shows exactly how such gcd mass can be amplified in integer determinants.

A second, logically independent route identifies the sparse logarithmic determinant with the numerical period of a symmetric tensor of Kummer logarithms. Its weight-$(1, 1)$ motivic coaction is precisely the formal prime symbol whose vanishing forces $M = 2^k$ and $N = 3^k$.

This yields a clean conditional proof under a restricted weight-two period-injectivity statement, and explains why the symmetric square, rather than the raw tensor square, is the correct receptacle. A function-field divisor theorem supplies an unconditional analogue and isolates the constant-field obstruction to specialization. The final sections give a focused experimental agenda and a Lean formalization plan for every unconditional component.

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1 Scope, status, and the nonduplication boundary

The problem is

$$2^x, 3^x \in \mathbb{Z} \qquad \Longrightarrow \qquad x \in \mathbb{Z}, \qquad x \in \mathbb{R}. \tag{1}$$

It originated in the work of Alaoglu and Erdős on highly composite and related integers [1]. In exponential form, a putative counterexample supplies the algebraic 2×2 array

$$\exp\begin{pmatrix} \log 2 & \log 3 \\ x\log 2 & x\log 3 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ M & N \end{pmatrix}, \qquad M, N \in \mathbb{Z}_{>0},$$

whose logarithmic matrix has rank one. This is a highly structured positive integral instance of the four-exponentials problem.

The supplied Report A already explores a large collection of determinant, fixed-prime p-adic, Mahler, interpolation, approximation, and logical routes. Its most useful common reduction is the free-monoid amplification: if $M = 2^x$ and $N = 3^x$, then

$$(2^a 3^b)^x = M^a N^b \in \mathbb{Z}_{>0} \qquad (a, b \in \mathbb{N}_0). \tag{2}$$

The present report imports only this compact baseline and does not repeat the prior surveys or determinant calculations. Its new project-level content is concentrated in four directions:

1. exact simultaneous p-orderings and generalized factorials of geometric progressions;
2. a rigorous half-mass and escape-of-mass theorem, including its full Vandermonde version;
3. a moving-prime gcd compatibility functional with a zero–one theorem and an explicit determinant amplifier;
4. a Kummer-period/coaction interpretation and a function-field divisor analogue.

The competitive claims mentioned in the task are not used as mathematical inputs. Every assertion below is either proved in the report, explicitly marked conditional, or attached to a primary mathematical reference.

1.1 Immediate reductions

Put

$$M = 2^x, \qquad N = 3^x.$$

If $x < 0$, then $0 < 2^x < 1$, so 2^x cannot be a positive integer. Thus $x \geq 0$. The case $x = 0$ is trivial.

Lemma 1.1 (Rational and algebraic cases)**.** *If $2^x \in \mathbb{Z}$ and $x \in \mathbb{Q}$, then $x \in \mathbb{Z}$. Consequently, a nonintegral counterexample to (1) must be transcendental.*

Proof. Write $x = a/b$ with $a, b \in \mathbb{N}$ and $\gcd(a, b) = 1$. If $M = 2^{a/b} \in \mathbb{Z}$, then $M^b = 2^a$. Taking the 2-adic valuation gives $b\, v_2(M) = a$, hence $b \mid a$ and $b = 1$. If x were algebraic irrational, the Gelfond–Schneider theorem would make 2^x transcendental; see, for example, [11, Ch. 1]. Therefore a nonintegral counterexample is transcendental. □

The exact real relation is the logarithmic determinant

$$\mathcal{D}(M, N) := \log 2 \, \log N - \log 3 \, \log M = 0. \tag{3}$$

The report will attack (3) in two different ways. The moving-prime route asks how much congruence structure is preserved by the monoid map in (2). The period route asks whether the numerical vanishing in (3) can be lifted to a formal relation between prime logarithm symbols.

2 Exact simultaneous p-orderings of geometric progressions

2.1 Definitions

Let $S \subset \mathbb{Z}$ and let p be prime. A sequence $s_0, s_1, s_2, \dots$ in S is a *p-ordering* if, at stage k, the quantity

$$v_p\left(\prod_{j=0}^{k-1}(s_k - s_j)\right)$$

is minimal among the corresponding quantities obtained by replacing s_k with an unused element of S. Bhargava introduced this construction and its associated generalized factorials in a much broader Dedekind-domain setting [3, 4]. A sequence that is a p-ordering for every prime simultaneously will be called a *simultaneous p-ordering*.

For integers $a, q \geq 1$ with $q \geq 2$, write

$$G(a,q) := \{aq^n : n \in \mathbb{N}_0\}.$$

The key observation is that the ordinary exponent ordering of $G(a,q)$ is simultaneously optimal at every prime. The proof is an exact Gaussian binomial identity, not an asymptotic estimate.

2.2 The Gaussian divisibility identity

For $0 \leq k \leq n$, define the Gaussian binomial coefficient by

$$\begin{bmatrix} n \\ k \end{bmatrix}_q := \prod_{r=1}^{k} \frac{q^{n-k+r}-1}{q^r - 1}. \tag{4}$$

It is an integer. One proof uses the recurrence

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k}\begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q$$

with the usual boundary conditions; see [2, Ch. 3].

Theorem 2.1 (Exact simultaneous ordering)**.** *Let $a, q \in \mathbb{Z}_{>0}$ and $q \geq 2$. The sequence*

$$a, aq, aq^2, aq^3, \dots$$

is a p-ordering of $G(a,q)$ for every prime p. More precisely, for all $n \geq k$,

$$\frac{\prod_{j=0}^{k-1}(aq^n - aq^j)}{\prod_{j=0}^{k-1}(aq^k - aq^j)} = \begin{bmatrix} n \\ k \end{bmatrix}_q \in \mathbb{Z}. \tag{5}$$

Consequently the kth generalized factorial is represented by

$$F_k(a,q) := \prod_{j=0}^{k-1}(aq^k - aq^j) = a^k q^{\binom{k}{2}} \prod_{r=1}^{k}(q^r - 1), \qquad F_0(a,q) := 1. \tag{6}$$

Proof. For $n \geq k$,

$$\prod_{j=0}^{k-1}(aq^n - aq^j) = a^k q^{0+1+\cdots+(k-1)} \prod_{j=0}^{k-1}(q^{n-j} - 1).$$

The corresponding expression with $n = k$ has the same powers of a and q. Their quotient is

$$\frac{\prod_{r=n-k+1}^{n}(q^r - 1)}{\prod_{r=1}^{k}(q^r - 1)} = \begin{bmatrix} n \\ k \end{bmatrix}_q,$$

which is an integer. Hence, for every prime p, the p-adic valuation at the candidate aq^n is at least the valuation at aq^k. Candidates with $n < k$ have already occurred and give a zero factor. This proves simultaneous minimality. Formula (6) follows by setting $n = k$ and reversing the product $k - j$. □

Remark 2.2 (Why simultaneity matters). A generic subset of $\mathbb{Z}$ has different greedy orderings at different primes. Here one ordering is globally compatible because every comparison quotient is an actual integer. This converts local p-adic information into a single integral denominator and permits exact global mass accounting.

2.3 A canonical Newton–Gaussian basis

Define

$$B_k^{(a,q)}(T) := \frac{\prod_{j=0}^{k-1}(T - aq^j)}{F_k(a, q)}. \tag{7}$$

Theorem 2.3 (Regular basis for the integer-valued polynomial ring)**.** *For every $n, k \in \mathbb{N}_0$,*

$$B_k^{(a,q)}(aq^n) = \begin{cases} 0, & n < k, \\ \begin{bmatrix} n \\ k \end{bmatrix}_q, & n \geq k. \end{cases} \tag{8}$$

In particular $B_k^{(a,q)}$ is integer-valued on $G(a, q)$ and $B_k^{(a,q)}(aq^k) = 1$. Every polynomial $P \in \mathbb{Q}[T]$ of degree at most d that is integer-valued on $G(a, q)$ has a unique expansion

$$P(T) = \sum_{k=0}^{d} c_k B_k^{(a,q)}(T), \qquad c_k \in \mathbb{Z}. \tag{9}$$

Proof. Equation (8) is exactly (5), with the zero case coming from a repeated node. For the expansion, determine c_k recursively from the values at $a, aq, \ldots, aq^d$. The evaluation matrix $(B_k^{(a,q)}(aq^n))_{0 \leq n,k \leq d}$ is lower triangular with ones on the diagonal and integral entries. Therefore the recursive coefficients are integers. The resulting polynomial agrees with P at $d+1$ distinct points, hence is equal to P. Uniqueness follows from the triangular matrix or from distinct degrees. □

This theorem supplies an exact denominator lattice for interpolation on a geometric progression. It is particularly relevant to (2): every ray

$$3^b,\ 3^b 2,\ 3^b 2^2, \ldots$$

and every ray

$$2^a,\ 2^a 3,\ 2^a 3^2, \ldots$$

inside the 2–3 smooth monoid has a canonical basis whose values and denominators are known exactly.

2.4 Exact Newton coefficients of a real power

The preceding basis also gives a useful identity for the nonpolynomial function $T \mapsto T^x$.

Proposition 2.4 (Real-power divided difference)**.** *Let $q > 1$, $x \in \mathbb{R}$, and put $Q = q^x$. For every $k \geq 0$,*

$$[1, q, \ldots, q^k]T^x = \frac{\prod_{r=0}^{k-1}(Q - q^r)}{F_k(1, q)}. \tag{10}$$

More generally,

$$F_k(a, q)\,[a, aq, \ldots, aq^k]T^x = a^x \prod_{r=0}^{k-1}(Q - q^r). \tag{11}$$

Here brackets denote ordinary divided differences.

Proof. As a function of the auxiliary variable $Y = Q$, the divided difference

$$C_k(Y) := \sum_{j=0}^{k} \frac{Y^j}{\prod_{0 \leq i \leq k,\ i \neq j}(q^j - q^i)}$$

is a polynomial of degree k. For $Y = q^r$ with $0 \leq r < k$, the values $Y^j = (q^j)^r$ come from a polynomial of degree r, so its kth divided difference is zero. Thus C_k has roots $1, q, \ldots, q^{k-1}$. Its leading coefficient comes only from the $j = k$ term and equals $1/F_k(1, q)$. This proves (10). Scaling every node by a multiplies a kth divided difference of the homogeneous function T^x by a^{x-k}; using $F_k(a, q) = a^k F_k(1, q)$ gives (11). □

For a putative Alaoglu–Erdős counterexample, (11) specializes to the two separated families

$$F_k(3^b, 2)\,[3^b, 3^b 2, \ldots, 3^b 2^k]T^x = N^b \prod_{r=0}^{k-1}(M - 2^r), \tag{12}$$

$$F_k(2^a, 3)\,[2^a, 2^a 3, \ldots, 2^a 3^k]T^x = M^a \prod_{r=0}^{k-1}(N - 3^r). \tag{13}$$

The right sides are integers. These identities do not by themselves prove that a divided difference is integral; instead, they show exactly where its denominator resides and how the transverse smooth parameter separates from the longitudinal geometric parameter.

Proposition 2.5 (Fixed-base denominator dichotomy)**.** *Let $p \mid q$, put $h = v_p(q)$, and suppose that $Q = q^x \in \mathbb{Z}_{>0}$ is not an integral power of q. If $s = v_p(Q)$, then*

$$v_p\left(\prod_{r=0}^{k-1}(Q - q^r)\right) = sk + O_{p,q,Q}(1), \tag{14}$$

and hence

$$v_p([1, q, \ldots, q^k]T^x) = -h\binom{k}{2} + sk + O_{p,q,Q}(1). \tag{15}$$

Proof. If $rh > s$, then

$$v_p(Q - q^r) = v_p(Q) = s.$$

If $rh < s$, then $v_p(Q - q^r) = rh$. There is at most one index with $rh = s$; its valuation is finite because $Q \neq q^r$. Thus all but finitely many terms have valuation s, proving (14). Since $p \mid q$ implies $p \nmid q^r - 1$, formula (6) gives $v_p(F_k(1, q)) = h\binom{k}{2}$. Subtracting valuations in (10) proves (15). □

The proposition is a rigorous warning: unless Q is already a q-power, the normalized divided differences acquire a quadratic denominator at the base primes, while their numerator can cancel only linearly. Any successful argument must therefore use arithmetic beyond the fixed primes dividing the base.

3 Exact local valuation profiles

The factorization in (6) permits a complete local calculation. This section records it because it distinguishes the fixed-base mass from the moving cyclotomic mass with no hidden constants.

Theorem 3.1 (Local valuation formula)**.** *Let $a, q \in \mathbb{Z}_{>0}$, $q \geq 2$, and $k \geq 1$.*

1. *If $p \mid q$, then*

$$v_p(F_k(a,q)) = kv_p(a) + \binom{k}{2} v_p(q). \tag{16}$$

2. *Let p be odd and $p \nmid q$. Put*

$$t = \operatorname{ord}_p(q), \qquad m = \left\lfloor \frac{k}{t} \right\rfloor, \qquad c = v_p(q^t - 1).$$

Then

$$v_p(F_k(a,q)) = kv_p(a) + mc + v_p(m!). \tag{17}$$

3. *If $p = 2$ and q is odd, put*

$$A = v_2(q-1), \qquad B = v_2(q+1), \qquad m = \lfloor k/2 \rfloor.$$

Then

$$v_2(F_k(a,q)) = kv_2(a) + kA + mB + v_2(m!). \tag{18}$$

Proof. The first assertion follows because $q^r - 1$ is a p-adic unit when $p \mid q$. For odd $p \nmid q$, the prime p divides $q^r - 1$ exactly when $t \mid r$. Writing $r = tu$, the lifting-the-exponent lemma gives

$$v_p(q^{tu} - 1) = v_p(q^t - 1) + v_p(u) = c + v_p(u).$$

Summing over $1 \leq u \leq m$ yields (17). If q is odd, the 2-adic lifting formula gives

$$v_2(q^r - 1) = \begin{cases} A, & r \text{ odd}, \\ A + B + v_2(r) - 1, & r \text{ even}. \end{cases}$$

For $r = 2u$ the second line is $A + B + v_2(u)$. Summing the odd and even indices produces $kA + mB + v_2(m!)$, proving (18). $\square$

Corollary 3.2 (Order-layer decomposition)**.** *For odd primes $p \nmid q$, all p-adic contribution to $F_k(1,q)$ is controlled by the residue order $t = \operatorname{ord}_p(q)$ and the single initial depth $v_p(q^t - 1)$. In particular, for fixed p,*

$$v_p(F_k(1,q)) = O_{p,q}(k).$$

For every fixed finite set of primes S,

$$\sum_{\substack{p \in S \\ p \nmid q}} v_p(F_k(1,q)) \log p = O_{q,S}(k). \tag{19}$$

The exact order-layer formula will reappear in the full Vandermonde setting, where the weight $k + 1 - r$ raises the scale from quadratic to cubic.

4 The half-mass law and escape from every fixed prime set

4.1 One-step generalized factorials

Set

$$\Phi_k(q) := \prod_{r=1}^{k}(q^r - 1), \qquad C_q := \sum_{r=1}^{\infty}\log(1-q^{-r}).$$

The series defining C_q converges absolutely.

Theorem 4.1 (Quadratic half-mass law)**.** *For fixed $q \geq 2$,*

$$\log F_k(1,q) = k^2\log q + C_q + O_q(q^{-k}), \tag{20}$$

$$\sum_{p\mid q} v_p(F_k(1,q))\log p = \binom{k}{2}\log q, \tag{21}$$

$$\log\Phi_k(q) = \binom{k+1}{2}\log q + C_q + O_q(q^{-k}). \tag{22}$$

Thus the primes dividing q carry asymptotically one half of the logarithmic mass of $F_k(1,q)$, and the cyclotomic factors $q^r - 1$ carry the other half. For fixed a, the extra term $k\log a$ is lower order and does not alter the limit.

Proof. From (6),

$$\begin{aligned}\log F_k(1,q) &= \binom{k}{2}\log q + \sum_{r=1}^{k}(r\log q + \log(1-q^{-r}))\\ &= k^2\log q + \sum_{r=1}^{k}\log(1-q^{-r}).\end{aligned}$$

The tail of the convergent series is $O_q(q^{-k})$. The base-prime identity follows from unique factorization of q, and the cyclotomic identity is the remaining factor. □

The asymptotic equality of the two halves might suggest that one can recover the cyclotomic half from a few favorable small primes. The next theorem shows that this is impossible.

Theorem 4.2 (Escape of cyclotomic mass)**.** *Fix $q \geq 2$ and a finite set of primes S. Then*

$$\sum_{\substack{p\notin S\\ p\nmid q}} v_p(F_k(1,q))\log p = \binom{k+1}{2}\log q + O_{q,S}(k). \tag{23}$$

Equivalently, all but $O_{q,S}(k)$ of the quadratic cyclotomic mass is carried by primes outside S.

Proof. The total cyclotomic mass is (22). The contribution from the fixed set S is $O_{q,S}(k)$ by Corollary 3.2. Subtraction proves the result. □

Corollary 4.3 (No finite-prime recovery of the missing half)**.** *If S contains every prime dividing q, then*

$$\sum_{p\in S} v_p(F_k(1,q))\log p = \frac{1}{2}\log F_k(1,q) + o(k^2).$$

No enlargement of S by finitely many primes changes the leading constant $1/2$.

4.2 Full geometric Vandermonde determinants

For the nodes $a, aq, \dots, aq^d$, define

$$V_d(a,q) := \prod_{0 \le i < j \le d} (aq^j - aq^i).$$

Theorem 4.4 (Exact geometric Vandermonde factorization)**.** *For $d \ge 1$,*

$$V_d(a,q) = a^{\binom{d+1}{2}} q^{\binom{d+1}{3}} \prod_{r=1}^{d} (q^r - 1)^{d+1-r}. \tag{24}$$

Moreover,

$$V_d(a,q) = \prod_{k=1}^{d} F_k(a,q). \tag{25}$$

Proof. Each difference is $aq^i(q^{j-i} - 1)$. There are $\binom{d+1}{2}$ factors of a, and

$$\sum_{0 \le i < j \le d} i = \sum_{i=0}^{d-1} i(d-i) = \binom{d+1}{3}.$$

A difference $r = j - i$ occurs $d + 1 - r$ times, proving (24). Grouping the pairs by their larger index $j = k$ gives (25). □

Theorem 4.5 (Cubic half-mass law)**.** *For fixed $q \ge 2$ and $a = 1$,*

$$\log V_d(1,q) = \frac{d(d+1)(2d+1)}{6} \log q + O_q(d), \tag{26}$$

$$\sum_{p \mid q} v_p(V_d(1,q)) \log p = \binom{d+1}{3} \log q, \tag{27}$$

$$\sum_{r=1}^{d} (d+1-r) \log(q^r - 1) = \binom{d+2}{3} \log q + O_q(d). \tag{28}$$

The fixed-base and cyclotomic terms are both asymptotic to $\frac{1}{6} d^3 \log q$, while the total is asymptotic to $\frac{1}{3} d^3 \log q$.

Proof. Only the error term needs comment. We have

$$\sum_{r=1}^{d} (d+1-r) \log(q^r - 1) = \log q \sum_{r=1}^{d} (d+1-r) r + \sum_{r=1}^{d} (d+1-r) \log(1 - q^{-r}).$$

The first sum is $\binom{d+2}{3}$. Since $\sum_r |\log(1 - q^{-r})| < \infty$, the second sum is $O_q(d)$. Combining this with (27) gives (26). □

Theorem 4.6 (Cubic escape of mass)**.** *Let $q \ge 2$ and let S be a fixed finite set of primes. Then*

$$\sum_{\substack{p \notin S \\ p \nmid q}} v_p(V_d(1,q)) \log p = \binom{d+2}{3} \log q + O_{q,S}(d^2). \tag{29}$$

In particular, if S contains the primes dividing q, then

$$\sum_{p \in S} v_p(V_d(1,q)) \log p = \binom{d+1}{3} \log q + O_{q,S}(d^2) = \left(\frac{1}{2} + o(1)\right) \log V_d(1,q). \tag{30}$$

Proof. For an odd fixed prime $p \nmid q$, let $t = \mathrm{ord}_p(q)$ and $c = v_p(q^t - 1)$. The local contribution to the cyclotomic part of (24) is exactly

$$\sum_{m=1}^{\lfloor d/t \rfloor} (d + 1 - tm)\big(c + v_p(m)\big), \tag{31}$$

by the lifting-the-exponent formula. This is $O_{p,q}(d^2)$. The analogous 2-adic expression, obtained from (18), is also $O_q(d^2)$. A finite sum over S is therefore $O_{q,S}(d^2)$. Subtracting it from the total cyclotomic mass (28) proves (29); adding the exact base contribution proves (30). □

Corollary 4.7 (Increasing residue orders)**.** *Fix $T \geq 1$. Among the cyclotomic prime factors contributing to $V_d(1, q)$, the primes with $\mathrm{ord}_p(q) \leq T$ contribute only $O_{q,T}(d^2)$. Consequently,*

$$\sum_{\substack{p \nmid q \\ \mathrm{ord}_p(q) > T}} v_p(V_d(1, q)) \log p = \binom{d+2}{3} \log q + O_{q,T}(d^2).$$

Thus asymptotically all of the cyclotomic half occurs at primes whose residue orders tend beyond every fixed cutoff.

Proof. The primes with $\mathrm{ord}_p(q) \leq T$ divide the fixed integer $\prod_{r=1}^{T}(q^r - 1)$, so they form a finite set. Apply Theorem 4.6. □

4.3 Interpretation for determinant arguments

The change-of-basis matrix from monomials $1, T, \ldots, T^d$ to the regular basis $B_0^{(a,q)}, \ldots, B_d^{(a,q)}$ is triangular. Its determinant is

$$\prod_{k=1}^{d} \frac{1}{F_k(a, q)} = \frac{1}{V_d(a, q)}.$$

Therefore $V_d(a, q)$ is not a convenient overestimate: it is the exact covolume of the canonical integer-valued interpolation lattice. The half-mass theorem has the following sharp methodological consequence.

Criterion 4.8 (Fixed-prime barrier)**.** *Suppose an auxiliary determinant on a q-geometric interpolation lattice requires, at leading order, the full covolume gain $\log V_d(1, q)$. Any valuation argument restricted to a fixed finite set of primes can recover at most*

$$\left(\frac{1}{2} + o(1)\right) \log V_d(1, q)$$

from the universal geometric denominator. Closing the remaining half requires primes whose residue orders and identities vary with d, or a non-p-adic mechanism of the same cubic strength.

This does not say that every fixed-prime proof is impossible: a sufficiently strong archimedean estimate might need less than the full covolume. It does say that simply sharpening constants in the same fixed-2/3 valuation bookkeeping cannot manufacture the missing cyclotomic half.

5 Moving-prime compatibility: a zero–one endgame

The half-mass theorem identifies where the missing arithmetic lives. We now measure whether the real-power monoid map preserves any of it.

5.1 The compatibility mass

For integers $q, Q > 1$, define

$$\mathfrak{C}_d(q,Q) := \sum_{r=1}^{d} (d+1-r) \log \gcd(q^r - 1, Q^r - 1) \tag{32}$$

and the total cyclotomic reference mass

$$\mathfrak{M}_d(q) := \sum_{r=1}^{d} (d+1-r) \log(q^r - 1). \tag{33}$$

By (28),

$$\mathfrak{M}_d(q) = \binom{d+2}{3} \log q + O_q(d). \tag{34}$$

The ratio $\mathfrak{C}_d/\mathfrak{M}_d$ records the weighted fraction of the moving cyclotomic mass of q that is also visible to Q.

The relevant external input is the theorem of Bugeaud–Corvaja–Zannier [5]: if q and Q are multiplicatively independent integers greater than one, then for every $\varepsilon > 0$,

$$\log \gcd(q^r - 1, Q^r - 1) \le \varepsilon r$$

for all sufficiently large r.

Theorem 5.1 (Compatibility zero–one law for a prime base)**.** *Let q be prime and $Q \in \mathbb{Z}_{>1}$.*

1. *If $Q = q^m$ for some $m \in \mathbb{N}$, then*

$$\mathfrak{C}_d(q,Q) = \mathfrak{M}_d(q) \tag{35}$$

for every d.

2. *If Q is not an integral power of q, then*

$$\mathfrak{C}_d(q,Q) = o(d^3), \qquad \frac{\mathfrak{C}_d(q,Q)}{\mathfrak{M}_d(q)} \longrightarrow 0. \tag{36}$$

Thus the normalized compatibility mass tends to either 1 *or* 0*; there is no intermediate positive limiting proportion.*

Proof. If $Q = q^m$, then $q^r - 1$ divides $q^{mr} - 1$, giving (35). Otherwise q and Q are multiplicatively independent. Indeed, a relation $q^u = Q^v$ and unique factorization would force Q to be a power of the prime q.

Fix $\varepsilon > 0$. By the Bugeaud–Corvaja–Zannier theorem there is R such that, for $r \ge R$,

$$\log \gcd(q^r - 1, Q^r - 1) \le \varepsilon r.$$

Hence

$$\begin{aligned} \mathfrak{C}_d(q,Q) &\le O_{q,Q,R}(d) + \varepsilon \sum_{r=R}^{d} (d+1-r) r \\ &\le O_{q,Q,R}(d) + \varepsilon \binom{d+2}{3}. \end{aligned}$$

After division by d^3, the limsup is at most $\varepsilon/6$. Since ε is arbitrary, (36) follows; the ratio statement uses (34). $\square$

Corollary 5.2 (A concrete sufficient condition for Alaoglu–Erdős)**.** *Let $x > 0$, and let $M = 2^x$ and $N = 3^x$ be integers. If*

$$\limsup_{d\to\infty} \frac{\mathfrak{C}_d(2, M) + \mathfrak{C}_d(3, N)}{d^3} > 0, \tag{37}$$

then $x \in \mathbb{Z}$.

Proof. Condition (37) implies that at least one of the two compatibility masses is not $o(d^3)$. By Theorem 5.1, either $M = 2^m$ or $N = 3^m$ for some $m \in \mathbb{N}$. In the first case $x = \log_2 M = m$; in the second, $x = \log_3 N = m$. □

Corollary 5.3 (Single-layer version)**.** *The same conclusion holds if, for one of the pairs $(q, Q) = (2, M)$ or $(3, N)$, there is a constant $c > 0$ and infinitely many r such that*

$$\log \gcd(q^r - 1, Q^r - 1) \geq cr.$$

The support theorem of Corrales–Rodríguez and Schoof [8] gives a related qualitative endpoint: sufficiently systematic transfer of prime supports or reduction orders from q to Q forces multiplicative dependence. The weighted functional (32) is adapted more directly to the geometric Vandermonde factorization, because the same weight $d + 1 - r$ occurs in (24).

5.2 The precise bridge conjecture

Conjecture 5.4 (Two-base moving-prime bridge)**.** *Let $M, N \in \mathbb{Z}_{>1}$ satisfy*

$$\frac{\log M}{\log 2} = \frac{\log N}{\log 3}.$$

Then

$$\limsup_{d\to\infty} \frac{\mathfrak{C}_d(2, M) + \mathfrak{C}_d(3, N)}{d^3} > 0.$$

Corollary 5.5. *Conjecture 5.4 implies the Alaoglu–Erdős conjecture.*

Proof. Apply Corollary 5.2. □

The bridge conjecture is deliberately narrow. It does not request full support transfer, equality of p-adic exponents, or continuity of the power map in every local topology. A positive cubic proportion in either pair is enough, and the zero–one theorem then upgrades that proportion to exact multiplicative dependence.

5.3 A determinant amplifier for gcd compatibility

The next lemma explains why the gcd in (32) is the correct local object for determinants built from the multiplicative semigroup $\langle q, Q\rangle$.

Lemma 5.6 (Block-Vandermonde amplification)**.** *Let $q, Q > 1$, let $r \geq 1$, and put*

$$g_r = \gcd(q^r - 1, Q^r - 1).$$

Choose distinct numbers

$$u_j = q^{a_j} Q^{b_j}, \qquad a_j, b_j \in \mathbb{N}_0, \qquad 0 \leq j \leq L.$$

For any $n_0 \in \mathbb{N}_0$, *define*

$$D = \det(u_j^{n_0+ir})_{0\le i,j\le L}.$$

Then

$$g_r^{\binom{L+1}{2}} \mid D. \tag{38}$$

More precisely,

$$D = \left(\prod_{j=0}^{L} u_j^{n_0}\right) \prod_{0\le i<j\le L} (u_j^r - u_i^r). \tag{39}$$

Proof. Equation (39) is the ordinary Vandermonde factorization applied to $u_0^r, \dots, u_L^r$. Modulo g_r one has $q^r \equiv Q^r \equiv 1$, hence $u_j^r \equiv 1$ for every j. Therefore every pairwise difference in (39) is divisible by g_r. There are $\binom{L+1}{2}$ such differences. □

There is an equivalent finite-difference proof: on the arithmetic progression $n_0, n_0+r, \dots, n_0+Lr$, each forward difference of a sequence $q^{an}Q^{bn}$ gains a factor g_r, and the ith transformed row gains g_r^i. The determinant amplification exponent is therefore $0 + 1 + \cdots + L$.

For the Alaoglu–Erdős data, one may use the same exponent pairs (a_j, b_j) in the two semigroups

$$u_j^{(2)} = 2^{a_j} M^{b_j} = 2^{a_j+b_j x}, \qquad u_j^{(3)} = 3^{a_j} N^{b_j} = 3^{a_j+b_j x}.$$

The real slopes $a_j + b_j x$ are identical while the local compatibility factors are

$$\gcd(2^r - 1, M^r - 1), \qquad \gcd(3^r - 1, N^r - 1).$$

This is the point at which the real common-exponent relation and the moving prime factors meet in one determinant architecture.

5.4 What remains to be proved in this route

A complete moving-prime proof would have to reverse the usual direction of a gcd estimate. Known theorems give strong *upper* bounds when the pairs are independent. The missing bridge must produce a *lower* bound from the simultaneous real relation. Three possible concrete formulations are:

1. prove Conjecture 5.4 directly;
2. produce infinitely many r satisfying the single-layer condition of Corollary 5.3;
3. construct a nonzero integer determinant whose divisibility lower bound contains a positive cubic portion of $\mathfrak{C}_d(2, M) + \mathfrak{C}_d(3, N)$, while the common real slope makes its archimedean size smaller than that divisor.

The half-mass theorem shows that the third formulation has enough available prime weight in principle. It also shows that a construction that ignores the varying factors $q^r - 1$ cannot reach the required scale.

6 A symmetric Kummer-period and coaction route

The moving-prime program is unconditional but lacks the bridge lower bound. The period route supplies a different kind of progress: it identifies the exact standard conjectural statement that would make the logarithmic determinant formal, after which unique factorization ends the proof.

6.1 The symmetric prime symbol

Let S be a finite set of primes containing all prime divisors of $6MN$, and let

$$E_S := \bigoplus_{p\in S} \mathbb{Q}e_p.$$

For a positive S-unit A, define its prime vector

$$\ell(A) := \sum_{p\in S} v_p(A)e_p.$$

Let $u \odot v = (u \otimes v + v \otimes u)/2$ and consider $\mathrm{Sym}^2(E_S)$. Define the numerical quadratic-log map

$$\pi_S : \mathrm{Sym}^2(E_S) \longrightarrow \mathbb{R}, \qquad \pi_S(e_p \odot e_q) = \log p \, \log q. \tag{40}$$

The relevant symbol is

$$\Xi(M,N) := e_2 \odot \ell(N) - e_3 \odot \ell(M). \tag{41}$$

Then

$$\pi_S(\Xi(M,N)) = \log 2 \, \log N - \log 3 \, \log M. \tag{42}$$

Theorem 6.1 (Formal prime normal form)**.** *For positive integers M, N,*

$$\Xi(M,N) = 0$$

if and only if there is $k \in \mathbb{N}_0$ such that

$$M = 2^k, \qquad N = 3^k.$$

Proof. Compare coefficients in the standard basis of $\mathrm{Sym}^2(E_S)$. For every $p \notin \{2,3\}$, the coefficient of $e_2 \odot e_p$ is $v_p(N)$, so $v_p(N) = 0$; the coefficient of $e_3 \odot e_p$ is $-v_p(M)$, so $v_p(M) = 0$. The coefficient of $e_2 \odot e_2$ is $v_2(N)$, and the coefficient of $e_3 \odot e_3$ is $-v_3(M)$, so these also vanish. Finally, the coefficient of $e_2 \odot e_3$ is

$$v_3(N) - v_2(M),$$

which must be zero. Thus $M = 2^k$ and $N = 3^k$ with a common k. The converse is immediate. □

Remark 6.2 (Why the symmetric square is essential)*.* The raw tensor $e_2 \otimes \ell(N) - e_3 \otimes \ell(M)$ does not vanish for the genuine solutions $M = 2^k$, $N = 3^k$: it becomes $k(e_2 \otimes e_3 - e_3 \otimes e_2)$. Numerical multiplication of logarithms kills this antisymmetric tensor because real multiplication is commutative. Passing to $\mathrm{Sym}^2(E_S)$ removes exactly this unavoidable kernel and no more.

6.2 Restricted period injectivity

Definition 6.3 (Quadratic prime-log injectivity)**.** For a finite prime set S, let $\mathrm{QPLI}(S)$ denote the assertion that the map π_S in (40) is injective. Equivalently, the products

$$\{\log p \, \log q : p, q \in S, \ p \le q\}$$

are linearly independent over $\mathbb{Q}$.

Theorem 6.4 (Conditional Alaoglu–Erdős theorem)**.** *Let $M = 2^x$ and $N = 3^x$ be positive integers, and let S contain the prime divisors of $6MN$. If $\mathrm{QPLI}(S)$ holds, then $x \in \mathbb{N}_0$.*

Proof. Equation (3) and (42) give $\pi_S(\Xi(M,N)) = 0$. Injectivity gives $\Xi(M,N) = 0$. By Theorem 6.1, $M = 2^k$ and $N = 3^k$ for some $k \in \mathbb{N}_0$. Therefore $x = \log M/\log 2 = k$. □

Schanuel's conjecture implies QPLI(S): the numbers $\log p$ are linearly independent over $\mathbb{Q}$ by unique factorization, and Schanuel then predicts their algebraic independence because their exponentials are algebraic. The point of the present formulation is not to replace one open problem by a larger one, but to isolate the exact weight-two subspace and its formal coefficient extraction.

6.3 Kummer periods and the weight-$(1,1)$ coaction

For $A \in \mathbb{Q}^{\times}_{>0}$, the logarithm $\log A$ is a period of a Kummer extension. In the category of mixed Tate motives over $\mathbb{Z}[S^{-1}]$, write schematically

$$L_A^{\mathfrak{m}} = \log^{\mathfrak{m}}(A), \qquad L_A^{\mathrm{dR}} = \log^{\mathrm{dR}}(A)$$

for the motivic and de Rham logarithm classes. The basic coaction is primitive:

$$\Delta L_A^{\mathfrak{m}} = L_A^{\mathfrak{m}} \otimes 1 + 1 \otimes L_A^{\mathrm{dR}}. \tag{43}$$

See Deligne–Goncharov [9] and Brown [6] for the mixed Tate and motivic-period frameworks. Multiplicativity of the coaction gives the weight-$(1,1)$ component

$$\Delta_{1,1}(L_A^{\mathfrak{m}} L_B^{\mathfrak{m}}) = L_A^{\mathfrak{m}} \otimes L_B^{\mathrm{dR}} + L_B^{\mathfrak{m}} \otimes L_A^{\mathrm{dR}}. \tag{44}$$

Define the motivic determinant period

$$\mathcal{D}^{\mathfrak{m}}(M,N) := L_2^{\mathfrak{m}} L_N^{\mathfrak{m}} - L_3^{\mathfrak{m}} L_M^{\mathfrak{m}}. \tag{45}$$

Its numerical period is exactly (3). Using $L_A = \sum_p v_p(A) L_p$, the symmetrization of $\Delta_{1,1}\mathcal{D}^{\mathfrak{m}}$ is the prime symbol $2\Xi(M,N)$.

Theorem 6.5 (Motivic conditional proof)**.** *Let S contain the prime divisors of $6MN$. Assume that the numerical period map is injective on the weight-at-most-two motivic-period subspace of the mixed Tate category generated by the Kummer classes $\{L_p^{\mathfrak{m}} : p \in S\}$. Then any real x with $2^x = M \in \mathbb{Z}$ and $3^x = N \in \mathbb{Z}$ is an integer.*

Proof. The numerical period of $\mathcal{D}^{\mathfrak{m}}(M,N)$ is zero, so the assumed injectivity gives $\mathcal{D}^{\mathfrak{m}}(M,N) = 0$. Applying the weight-$(1,1)$ coaction and using (44), then identifying the independent prime Kummer classes in the motivic and de Rham factors, gives $\Xi(M,N) = 0$. The conclusion follows from Theorem 6.1. □

The hypothesis is a restricted instance of the expected injectivity of the period map in the Grothendieck–Kontsevich–Zagier philosophy [10]. The new structural point is the explicit coaction computation: the project-level formal prime symbol is not merely an analogy. It is the weight-$(1,1)$ coaction shadow of the tensor-square Kummer period whose numerical value is the logarithmic determinant.

6.4 What the coaction does and does not solve

The coaction extracts prime coefficients only after a numerical relation has been lifted to a motivic relation. That lifting is the unproved step. It would be circular to declare numerical vanishing sufficient without a period injectivity theorem. Nevertheless, the route gives three useful pieces of information:

1. it identifies the correct quotient (the symmetric square) and removes the false antisymmetric obstruction;
2. it reduces every genuine solution to a coefficient comparison in weight one;
3. it suggests that a moving-prime determinant should be viewed as an unconditional arithmetic surrogate for the coefficient-extracting coaction.

The last point links the two main routes: the coaction sees all prime symbols at once, while the half-mass theorem says that an elementary determinant must let the relevant primes move with the scale to emulate that global coefficient extraction.

7 A function-field divisor analogue and the constant-field obstruction

In a function field, valuations provide the missing coefficient detector unconditionally. This makes the function-field analogue easy and shows precisely what is lost when the bases specialize to constants.

7.1 Divisor content

Let C be a smooth projective geometrically integral curve over a field K, and let $K(C)$ be its function field. For $F \in K(C)^\times$ define

$$c(F) := \gcd\{|v_P(F)| : P \in C,\ v_P(F) \neq 0\}. \tag{46}$$

If F is constant, the set is empty; below we apply the definition only to nonconstant functions.

Theorem 7.1 (Function-field exponent rigidity)**.** *Let $F, G, H, J \in K(C)^\times$ with F and G nonconstant, and let $x \in \mathbb{R}$. Suppose that the equalities of real divisors*

$$\operatorname{div}(H) = x \operatorname{div}(F), \qquad \operatorname{div}(J) = x \operatorname{div}(G) \tag{47}$$

hold. Then $x \in \mathbb{Q}$. If $x = a/b$ in lowest terms, then

$$b \mid c(F), \qquad b \mid c(G).$$

In particular, if $\gcd(c(F), c(G)) = 1$, then $x \in \mathbb{Z}$.

Proof. Choose a point P with $v_P(F) \neq 0$. From the first equality in (47),

$$x = \frac{v_P(H)}{v_P(F)} \in \mathbb{Q}.$$

Write $x = a/b$ in lowest terms. For every P, $b\,v_P(H) = a\,v_P(F)$; since $\gcd(a, b) = 1$, one has $b \mid v_P(F)$. Thus $b \mid c(F)$. The same argument applies to G. If the two contents are coprime, then $b = 1$. □

The criterion is sharp: if $F = U^b$ and $H = U^a$, then $\operatorname{div}(H) = (a/b)\operatorname{div}(F)$. For the natural deformation of the bases,

$$F(t) = t, \qquad G(t) = t + 1$$

on $\mathbb{P}^1$, both divisor contents are one, so any rational-function lift of t^x and $(t+1)^x$ would force $x \in \mathbb{Z}$.

7.2 Why specialization to 2 and 3 loses the proof

Lemma 7.2 (Constant-field kernel). *If C is projective and geometrically integral, then*

$$\operatorname{div}(F) = 0 \qquad \Longleftrightarrow \qquad F \in K^\times.$$

Proof. A rational function with no zeros or poles is a global invertible regular function on a proper integral curve, hence is constant. □

At the special value $t = 2$, the functions t and $t + 1$ become the constants 2 and 3. Their divisors vanish, and the coefficient detector in Theorem 7.1 disappears entirely. A deformation proof must therefore supply more than a family passing through the four numbers $(2, 3, M, N)$. It must lift the exponent relation to nonconstant rational or motivic objects while controlling divisor content. The equalities at a single constant fiber do not produce such a lift.

This is the same obstruction seen in the period route. Divisors make prime coefficients visible over a function field; motivic coactions are expected to make prime Kummer coefficients visible over the constant field; moving-prime Vandermonde factors attempt to recover part of that visibility by elementary congruences.

8 Synthesis: a focused proof program

The new results suggest a much narrower research plan than another broad search through unrelated transcendence techniques.

8.1 Stage I: build determinants that spend the cyclotomic half

The geometric factorial and Vandermonde formulas specify the available local budget exactly. A candidate determinant should satisfy all of the following.

1. Its interpolation nodes are organized into 2- and 3-geometric blocks, or into blocks in the semigroups $\langle 2, M\rangle$ and $\langle 3, N\rangle$.
2. Its row operations expose factors $q^r - 1$ with the natural weight $d + 1 - r$, rather than using only the base-prime factors $q^{\binom{d+1}{3}}$.
3. The nonpolynomial columns created by the exponent x are paired so that their loss of congruence is measured by $\gcd(q^r - 1, Q^r - 1)$.
4. The common slope $\log M/\log 2 = \log N/\log 3$ cancels the leading archimedean growth between the two base blocks.

The block amplifier in Lemma 5.6 is a basic module for such a construction.

8.2 Stage II: prove a positive compatibility lower bound

The decisive target is no longer an unspecified "extra p-adic gain." It is one of the explicit inequalities

$$\mathfrak{C}_d(2, M) + \mathfrak{C}_d(3, N) \gg d^3 \quad \text{along a subsequence,} \tag{48}$$
$$\log \gcd(2^r - 1, M^r - 1) + \log \gcd(3^r - 1, N^r - 1) \gg r \quad \text{for infinitely many } r. \tag{49}$$

Either statement closes the conjecture by Theorem 5.1 and Corollary 5.3. The weighted version (48) is naturally aligned with the exact Vandermonde covolume; the single-layer version (49) is more rigid and may be easier to connect to support-problem methods.

8.3 Stage III: use the period route as a structural benchmark

The Kummer coaction says what a perfect coefficient detector would output:

$$e_2 \odot \ell(N) - e_3 \odot \ell(M).$$

A determinant construction can be evaluated against this benchmark. If its moving-prime divisibility only sees aggregate heights and not the separate prime coefficients, it is unlikely to distinguish a true integer exponent from an unrelated multiplicatively independent pair. The desired determinant should approximate coefficient extraction by stratifying primes according to $\mathrm{ord}_p(q)$ and by comparing which strata survive under $q \mapsto Q$.

8.4 A precise theorem stack

A complete proof along the moving-prime route can be decomposed into the following stack.

Layer	Statement
A	Exact simultaneous p-ordering and denominator lattice on every geometric ray (proved in Theorems 2.1 and 2.3).
B	Half-mass and escape-of-mass at quadratic and cubic scale (proved in Theorems 4.1, 4.2, 4.5 and 4.6).
C	Compatibility zero–one law from gcd estimates (proved in Theorem 5.1).
D	Determinant amplification of a given compatibility factor (proved in Lemma 5.6).
E	Common-real-slope $\Rightarrow$ positive moving-prime compatibility (open bridge, Conjecture 5.4).
F	Positive compatibility $\Rightarrow x \in \mathbb{Z}$ (proved in Corollary 5.2).

Only Layer E is missing from this stack.

9 Sanity checks and small exact examples

The exact formulas are simple enough to audit by hand.

9.1 Generalized factorials

For $q = 2$ and $a = 1$,

$$F_1 = 1, \qquad F_2 = 6, \qquad F_3 = 168, \qquad F_4 = 20160.$$

Indeed,

$$F_4 = 2^{\binom{4}{2}}(2-1)(2^2-1)(2^3-1)(2^4-1) = 2^6 \cdot 1 \cdot 3 \cdot 7 \cdot 15.$$

For $q = 3$,

$$F_1 = 2, \qquad F_2 = 48, \qquad F_3 = 11232.$$

9.2 Vandermonde products

For the nodes $1, 2, 4, 8$,

$$V_3(1,2) = (2-1)(4-1)(8-1)(4-2)(8-2)(8-4) = 1008.$$

The product-of-factorials identity gives the same value:

$$F_1 F_2 F_3 = 1 \cdot 6 \cdot 168 = 1008.$$

The factorization (24) gives

$$2^{\binom{4}{3}}(2-1)^3(2^2-1)^2(2^3-1) = 2^4 \cdot 1^3 \cdot 3^2 \cdot 7 = 1008.$$

For the nodes $1, 3, 9$,

$$V_2(1,3) = 2 \cdot 8 \cdot 6 = 96 = F_1(1,3)F_2(1,3).$$

9.3 Newton coefficient check

For $k = 2$,

$$[1, q, q^2]T^x = \frac{(q^x-1)(q^x-q)}{q(q-1)(q^2-1)}.$$

At $x = 0$ or $x = 1$ the numerator vanishes, and at $x = 2$ it equals the denominator, as required for the polynomial T^2. If $q^x = Q$ is an integer but not a power of q, the numerator is a nonzero integer and Proposition 2.5 predicts a growing base-prime denominator as k increases.

9.4 Compatibility extremes

If $Q = q^m$, then

$$\gcd(q^r - 1, Q^r - 1) = q^r - 1$$

for every r, so the compatibility mass is full. At the opposite extreme, for multiplicatively independent q, Q, the Bugeaud–Corvaja–Zannier theorem forces the normalized weighted mass to zero. This checks that the functional (32) sharply distinguishes the desired conclusion rather than merely measuring generic common small primes.

10 Lean formalization plan

The unconditional core is well suited to formalization because its main identities are finite products and valuations. The motivic section should be formalized only as an abstract conditional interface until a suitable motive library exists.

10.1 Suggested module decomposition

File	Content
`GeomOrdering/Basic.lean`	Definitions of $G(a, q)$, the difference product $F_k(a, q)$, and the exact factorization in (6).
`GeomOrdering/Gaussian.lean`	Recursive definition or polynomial evaluation of Gaussian binomial coefficients; proof that the quotient in (5) is an integer.
`GeomOrdering/POrdering.lean`	Simultaneous p-ordering theorem and the regular Newton–Gaussian basis.
`GeomOrdering/Valuation.lean`	The three cases in Theorem 3.1, with odd-prime and 2-adic lifting lemmas isolated.
`GeomOrdering/Mass.lean`	Finite-product logarithm identities, convergence of C_q, and quadratic and cubic asymptotics.
`GeomOrdering/Vandermonde.lean`	Formula (24), product of generalized factorials, and fixed-finite-set escape estimates.
`Compatibility/WeightedGCD.lean`	Definitions of $\mathfrak{C}_d$ and $\mathfrak{M}_d$, the formal reduction from the BCZ upper-bound interface to the zero–one law, and Corollary 5.2.
`Compatibility/BlockVandermonde.lean`	Determinant factorization and divisibility in Lemma 5.6.
`PrimeSymbol/Symmetric.lean`	Finitely supported prime-exponent vectors, symmetric tensors, and the coefficient proof of Theorem 6.1.
`FunctionField/ExponentDivisor.lean`	Divisor content and Theorem 7.1, using the valuation API for function fields.
`Motivic/AbstractBridge.lean`	An abstract graded commutative period algebra with a primitive coaction; theorem that period injectivity plus the coaction implies the prime-symbol relation.

10.2 Core theorem sketches

The first theorem can be represented without introducing p-orderings initially:

```
theorem geom_diff_prod_dvd
    (a q n k : N) (hq : 2 <= q) (hk : k <= n) :
    product j in range k, (a*q^k - a*q^j)
      | product j in range k, (a*q^n - a*q^j)
```

After proving the quotient is the evaluated Gaussian binomial coefficient, p-ordering minimality is a one-line valuation consequence.

The weighted gcd theorem should isolate the deep external theorem behind a clean interface:

```
def BCZGcdBound (q Q : N) : Prop :=
  forall eps > 0, exists R,
    forall r >= R,
      log (gcd (q^r - 1) (Q^r - 1)) <= eps * r
```

The formal proof that this interface implies $\mathfrak{C}_d(q, Q) = o(d^3)$ is elementary finite-sum analysis. In this way, formalization can certify the entire reduction even before BCZ itself is formalized.

10.3 Dependency order

A practical order is

$$\text{finite products} \longrightarrow q\text{-binomial divisibility} \longrightarrow p\text{-ordering} \longrightarrow \text{local valuations},$$
$$\text{local valuations} \longrightarrow \text{mass laws} \longrightarrow \text{compatibility reduction}.$$

The prime-symbol normal form is independent and can be completed in parallel. The function-field theorem is also independent. This separation makes it possible to obtain a machine-checked package of all unconditional progress without waiting for the open bridge conjecture.

11 Failure modes and falsification checklist

A research report on an open problem is useful only if it states where its new routes can fail.

11.1 Moving-prime route

1. **Availability is not transfer.** The half-mass theorem proves that many moving primes divide geometric denominators. It does not prove that the map $q^n \mapsto Q^n$ preserves any of them.
2. **Determinants usually give upper bounds for gcds.** The divisibility in Lemma 5.6 turns a known gcd into a large divisor. A proof still needs the common real slope to force a determinant small enough, or otherwise force the gcd large enough.
3. **Primitive divisors alone are insufficient.** Zsigmondy's theorem [12] supplies new prime support for most $q^r - 1$, but a single unweighted prime per layer need not carry cubic logarithmic mass. The weighted mass, not merely the count of new primes, must enter.
4. **A fixed set cannot simulate motion.** Adding any finite list of auxiliary primes changes only the $O(d^2)$ error in (30).

11.2 Period route

1. **Numerical zero is not motivic zero unconditionally.** The coaction can be applied to a motivic relation, not directly to an unexplained real-number equality.
2. **The raw tensor is the wrong object.** It incorrectly rejects the genuine integer solutions; symmetrization is mandatory.
3. **The restricted injectivity is still deep.** It is a precise weight-two period statement, not an established theorem for arbitrary prime sets.

11.3 Deformation route

1. A family whose generic fiber has primitive divisors would solve the exponent problem, but the constant fiber $(2, 3)$ has zero divisor.
2. Interpolating finitely many values does not produce a rational or motivic lift of t^x.
3. Any claimed specialization argument must explicitly control divisor content through the degeneration; otherwise it silently assumes the missing bridge.

12 Conclusions

The report establishes the following unconditional progress.

1. The natural ordering of every integral geometric progression is a simultaneous p-ordering, with exact generalized factorial
$$F_k(a,q) = a^k q^{\binom{k}{2}} \prod_{r=1}^{k} (q^r - 1).$$
2. The corresponding Newton–Gaussian polynomials form a canonical integral basis for all rational polynomials integer-valued on the progression.
3. Fixed base primes carry exactly one half of the leading denominator mass, both at the single-factorial scale k^2 and at the full Vandermonde scale d^3.
4. The other half escapes every fixed finite prime set and occurs at residue orders tending beyond every fixed bound.
5. The weighted compatibility mass $\mathfrak{C}_d(q,Q)$ obeys a zero–one law for prime q: full mass for an integral power and $o(d^3)$ for multiplicative independence.
6. A positive combined compatibility mass for $(2,M)$ and $(3,N)$ is an unconditional sufficient condition for the Alaoglu–Erdős conclusion.
7. Block Vandermonde determinants amplify each compatibility factor by a quadratic exponent, giving a concrete mechanism for spending moving prime mass.
8. The logarithmic determinant is the period of a symmetric tensor of Kummer logarithms; its weight-$(1,1)$ coaction is exactly the formal prime symbol whose vanishing forces $M = 2^k$, $N = 3^k$.
9. A function-field divisor theorem proves the analogous exponent rigidity and isolates the constant-field kernel that blocks naive specialization.

No unconditional proof of (1) is claimed. The main open task has, however, been reduced to a sharply testable statement rather than a vague request for stronger estimates: prove that a common real exponent forces a positive amount of moving cyclotomic compatibility in at least one of the pairs $(2,M)$ and $(3,N)$. The exact p-ordering and mass formulas show that the required arithmetic budget exists; the BCZ zero–one theorem shows that any positive cubic fraction would immediately force the desired integral power; and the Kummer coaction identifies the formal coefficient relation that an ideal determinant should emulate.

A Two useful exact identities

A.1 Weighted local valuation in closed form

For an odd prime $p \nmid q$, let $t = \mathrm{ord}_p(q)$, $c = v_p(q^t - 1)$, and $m = \lfloor d/t \rfloor$. From (24) and lifting the exponent,
$$v_p(V_d(1,q)) = \sum_{r=1}^{d} (d+1-r) v_p(q^r - 1) \tag{50}$$
$$= \sum_{u=1}^{m} (d+1-tu)(c + v_p(u)). \tag{51}$$

This is the exact finite formula behind the $O(d^2)$ estimate for each fixed cyclotomic prime. It also suggests a computational stratification by residue order t rather than by prime size.

A.2 Asymptotic Newton fingerprint

If $Q = q^x$ and $x \notin \mathbb{N}_0$, then

$$(-1)^k q^{-\binom{k}{2}} \prod_{r=0}^{k-1} (Q - q^r) \longrightarrow \prod_{r=0}^{\infty} (1 - q^{x-r}), \tag{52}$$

where the infinite product converges to a nonzero real number. Indeed,

$$Q - q^r = -q^r(1 - q^{x-r}),$$

and $\sum_r |q^{x-r}| < \infty$ in the tail. If $x = m \in \mathbb{N}_0$, the product is identically zero for $k > m$. Thus the Newton coefficient sequence has a sharp zero/nonzero dichotomy, although its nonzero asymptotic by itself does not supply the missing arithmetic contradiction.

B Minimal experimental agenda

The formulas in the report suggest computations that are diagnostic rather than merely numerical.

1. For candidate integer pairs (M, N) with $\log M/\log 2 \approx \log N/\log 3$, compute the finite compatibility profiles

   $$r \longmapsto \big(\log \gcd(2^r - 1, M^r - 1), \log \gcd(3^r - 1, N^r - 1)\big).$$

   Approximate pairs should display the BCZ-independent regime; any anomalous mass concentration identifies the residue-order layers a determinant must exploit.
2. Group the prime factors of $q^r - 1$ by $\mathrm{ord}_p(q) = r$ and compare their survival under $Q^r - 1$. The weighted quantity in (32) should be tracked, not only the number of common primes.
3. Search over lattice exponent pairs (a_j, b_j) in Lemma 5.6. Optimize the ratio between the guaranteed divisor $g_r^{\binom{L+1}{2}}$ and the exact archimedean determinant size under the common slope $a_j + b_j x$.
4. Test multiscale blocks with several step sizes r. Because valuations are assessed prime by prime, different residue orders can use different row orderings; a successful construction need not expose all moving primes in one literal sequence of row operations.

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